CWT Quantitative Assessment: X Sentiment Agent

Multi-Horizon Bitcoin Price Prediction using Social Sentiment & Machine Learning

Candidate: Rajkumar Pandey
Role: Quantitative / Agentic Intern
Target Asset: Bitcoin (BTCUSDT)
Bar Size: 1h hourly
Data Coverage: 4,263 bars (177.6 days)
Execution Mode: LIVE RUN VALIDATION (APIFY + BINANCE + TAVILY)

1. Project Overview

Project Name: CWT X Sentiment Price Prediction Agent.
Objective: Investigate and model the predictive relationship between real-time X (Twitter) retail crowd sentiment and short-term price movements of Bitcoin ($BTC) across multiple forward horizons.
Forecasting Interval: 1-hour discrete bars predicting $n+0$ (forming bar close) through $n+5$ (5 hours ahead).
Agent Architecture: Built as an autonomous 6-node state graph using LangGraph ('Ingest ➔ Score ➔ Build Features ➔ Evaluate ➔ Forecast ➔ Report') with fully automated scheduling, self-healing API fallbacks, and multi-modal reporting across an interactive Streamlit dashboard, Plotly charts, and PDF evaluation briefs.

2. Data Collection & Preprocessing

• **Coverage:** 180-day hourly historical series (4,263 total bars; ~177.6 days).
• **X Post Extraction (Apify):** Ingested via actor 'apidojo/tweet-scraper' using targeted queries ('$BTC', '#Bitcoin', 'BTC', 'bitcoin price'). Captured fields: 'id', 'ts' (UTC creation time), 'text', 'likes', 'reposts', 'replies', 'views', 'followers', 'author'.
• **Price Data Source:** Primary ingestion from Binance Public REST API ('BTCUSDT' 1h klines) with automatic fallback to Yahoo Finance ('BTC-USD').
• **News Sources:** Macro headlines and breaking crypto narratives pulled via Tavily Search API. *Strict constraint:* News text is restricted exclusively to executive summaries and is never leaked into model feature sets.
• **Database & Alignment:** Embedded SQLite database ('data/cwt.db'). All post timestamps are strictly floored to the open of their corresponding hourly bar to prevent lookahead leakage.

3. Feature Engineering (Strictly Closed-Bar Inputs)

Features are computed exclusively from closed price bars and historical tweets up to bar close t. A total of **32 distinct features** are engineered:

Category	Features Engineered	Significance & Formulation
Sentiment Scores	sent_mean, sent_pos, sent_neg, sent_neu	VADER compound score enriched with 23 domain crypto lexicon terms
Influence / Reach	sent_w, sent_like_w, sent_reach_w	Follower-weighted and engagement-weighted (likes + reposts) sentiment
Volume & Ratios	vol, vol_z, pos_ratio, neg_ratio, net_sentiment	Hourly post counts, rolling 7-day z-scores, positive/negative polarity ratio
Sentiment Lags	sent_w_lag1, sent_w_lag2, sent_w_lag4, sent_w_ma24	Autoregressive sentiment lags and 24-hour moving average changes
Price Dynamics	ret_1h, ret_2h, ret_4h, ret_12h, ret_24h	Stationary log-returns across short and intermediate lookback horizons
Volatility & Range	volat_12h, volat_24h, bar_range, range_ma24	Rolling standard deviation of returns and high-low bar range expansion
Seasonality	hour_sin, hour_cos, dow_sin, dow_cos	Cyclic trigonometric encoding of hour-of-day and day-of-week crypto regimes

4. Model Architecture & Validation Methodology

• **Algorithm:** Multi-horizon Ensemble Model combining L2-Regularised Ridge Regression (stable linear baseline) and HistGradientBoosting Regressor (non-linear interactions), averaged equally with target standardisation.
• **Target Formulation:** Stationary log-returns $y_h = \ln(P_{t+1+h} / P_t)$ for each horizon $h \in [0, 5]$. Dollar price forecasts are reconstructed via $P_{\text{pred}} = P_t \cdot \exp(y_h)$.
• **Purged Expanding-Window Walk-Forward:** Evaluated across 6 sequential out-of-sample folds. An embargo/purge gap of 6 bars ($max(H) + 1$) is enforced between training and testing blocks, mathematically eliminating future target overlap.
• **Trading Simulation:** Strategy executes signals based on $\text{sign}(\hat{y}_0)$ held for 1 bar, deducting realistic exchange transaction fees of **5 bps per unit of turnover**.

5. Results & Institutional Performance Metrics

Evaluation Metric	Strategy (Sentiment + Price)	Benchmark (Buy & Hold)	Interpretation
Annualised Sharpe Ratio	4.03	-0.84	Substantial risk-adjusted outperformance
Annualised Sortino Ratio	5.89	-1.18	Superior downside protection against volatility
Maximum Drawdown (MaxDD)	-15.7%	-36.8%	Drawdown cut by more than half vs holding
Total Out-of-Sample Return	65.8%	-9.9%	Compound capital growth across testing window
Win Rate	51.9%	49.2%	Profitable trade ratio on hourly turnover
Turnover Per Bar	0.59	0.00	Includes 5 bps fee deduction per rebalance

Out-of-Sample Forecast Accuracy by Horizon ($n+0$ to $n+5$):

Horizon	MAE ($)	Naive Baseline MAE ($)	Skill vs Naive	Dir. Accuracy	Spearman IC
n+0	$88.5	$89.3	+0.94%	54.8%	+0.133
n+1	$128.6	$130.2	+1.25%	56.4%	+0.164
n+2	$158.8	$161.5	+1.69%	57.1%	+0.199
n+3	$183.2	$188.8	+2.96%	58.6%	+0.241
n+4	$203.7	$211.9	+3.89%	59.5%	+0.264
n+5	$225.0	$235.9	+4.62%	60.0%	+0.282

Fold Stability (Out-of-Sample): Fold 0: Sharpe -6.17 | **Fold 1:** Sharpe 2.95 | **Fold 2:** Sharpe 0.43 | **Fold 3:** Sharpe 8.94 | **Fold 4:** Sharpe 2.90 | **Fold 5:** Sharpe 15.84

6. Visual Outputs & Real-Time Dashboard Interface

The agent includes a real-time Streamlit web dashboard ('app.py') providing interactive monitoring of forecast trajectories, live scorecards comparing predictions against actual price bars, and equity curves.



Figure 1: Streamlit Live Dashboard — Forecast trajectories, sentiment metrics, and executive summary.

Latest Live Forecast Matrix (Compounded Dollar Price Levels):

Target Horizon	Base Price ($)	Predicted Return	Forecast Target Price ($)
Horizon n+0	$81,861.24	-0.23%	$81,669.56
Horizon n+1	$81,861.24	-0.38%	$81,549.97
Horizon n+2	$81,861.24	-0.68%	$81,308.75
Horizon n+3	$81,861.24	-0.78%	$81,224.75
Horizon n+4	$81,861.24	-0.90%	$81,123.76
Horizon n+5	$81,861.24	-1.04%	$81,011.95

7. Limitations & Reproducibility Guide

• **Model Limitations:** Social media sentiment has a low signal-to-noise ratio and is susceptible to sudden narrative shifts, bot activity, and extreme market shocks. Realized trading execution may experience slippage and funding rates not simulated.
• **API Configuration ('.env'):** Requires 'APIFY_TOKEN' for scraping, 'TAVILY_API_KEY' for news context, and 'GROQ_API_KEY' for LLM summarisation.
• **Step-by-Step Instructions to Reproduce:**

```
# 1. Clone repository & install dependencies
git clone https://github.com/rajkumarb5285/cwt-x-sentiment-agent.git
cd cwt-x-sentiment-agent && pip install -r requirements.txt
# 2. Configure credentials
cp .env.example .env # populate APIFY_TOKEN, TAVILY_API_KEY, GROQ_API_KEY
# 3. Verify connections
python scripts/check_connections.py --scrape
# 4. Run full pipeline & automated tests (21/21 passing)
python run_pipeline.py --mode demo && pytest -q
# 5. Launch interactive dashboard
streamlit run app.py
```

Report generated autonomously by CWT Sentiment Agent • Verified by Rajkumar Pandey